

ASWIN P T

AI/ML Engineer · MSc Computer Science (Machine Intelligence)

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PROFESSIONAL SUMMARY

Machine Intelligence graduate student at Digital University Kerala with hands-on experience in federated learning, graph neural networks, differential privacy, and end-to-end AI engineering. Completed a 12-week industry internship delivering privacy-preserving AI systems for healthcare and financial fraud detection. Skilled in Python, PyTorch, FastAPI, and GPU-accelerated model development. Driven by research rigour, production-grade engineering discipline, and a strong commitment to building AI systems that are both performant and privacy-aware.

EDUCATION

M.Sc. Computer Science (Machine Intelligence) 2024 – 2026

Kerala University of Digital Sciences, Innovation and Technology (Digital University Kerala), Thiruvananthapuram

B.Sc. Computer Science 2021 – 2024

College of Applied Science IHRD, Kozhikode

TECHNICAL SKILLS

Programming: Python, C, C++

AI / ML: Deep Learning, Federated Learning, Graph Neural Networks, Differential Privacy, Computer Vision, Cognitive Computing, Data Analytics

Frameworks: PyTorch, TensorFlow, OpenCV, Flower (FL), Opacus (DP-SGD), FastAPI

Tools: Google Colab, Kaggle, VS Code, PyCharm, Jupyter Notebook, Docker, GitHub, Google Cloud Platform

Databases: PostgreSQL, SQL

Web: HTML, CSS, JavaScript, Django

INTERNSHIP EXPERIENCE

AI/ML Intern : Synbrains Solutions Pvt Ltd Mar 2026 – May 2026

12-week progressive internship covering production AI engineering, backend development, federated learning, and privacy-preserving ML across healthcare and financial domains.

PI-RADS Prostate AI Reporting System

- Contributed to a clinical AI reporting platform updated report layouts, implemented dropdown improvements, completed scoring, image handling, and sector mapping modules.
- Applied FastAPI backend integration and Git branching workflows for collaborative feature development.

Shear Wave Elastography Report Parser

- Developed an automated medical report extraction pipeline from specification documents with zero prior codebase.
- Built structured extraction pipelines using Pydantic data contracts and heuristic regex-based parsing to normalise raw PDF/DOCX report content into validated ReportDraft schemas.

PROJECTS

FedGraph-Shield : Hierarchical Federated Graph Learning for Financial Fraud Detection Ongoing

- Designed a hierarchical federated graph learning framework allowing distributed banking institutions to collaboratively train a fraud detection model without sharing raw transaction data.
- Modelled financial transactions as directed heterogeneous graphs customers, merchants, and devices as nodes; payments as edges enabling detection of ring structures and circular payment chains invisible to tabular approaches.
- Coordinated federated aggregation via the Flower framework using a two-tier hierarchical FedAvg strategy with differential privacy composition for formal privacy accounting across communication rounds.

AI-Enhanced PREDICT-ASD Early ASD Screening Platform

Completed

Python · Django · Hybrid AI (Rule-Based + ML) · React · PostgreSQL

- Built a full-stack, role-based web platform for structured ASD behavioural screening, interactive game-based evaluations, real-time caregiver–clinician communication, and automated report generation.
- Implemented a hybrid intelligence engine combining rule-based clinical logic with data-driven analysis to produce explainable preliminary ASD risk categorization.
- Cloud-deployed for scalability and centralised data management; validated through system testing demonstrating digitised screening workflows and improved caregiver participation.

Dynamic Connectome Based Prediction of Reality Monitoring Failure Under Sleep Deprivation (Graph Attention Networks)

Ongoing – Preprocessing Phase

Python · PyTorch · fMRI Preprocessing · Graph Attention Networks

- Designing a dynamic connectome framework using resting-state fMRI to predict reality monitoring failure before overt behavioural symptoms appear with applications to psychosis-risk and fatigue research.
- Constructing time-resolved functional connectivity matrices via a sliding-window approach; each matrix represented as a graph (brain regions = nodes, functional connections = weighted edges).
- Employing Graph Attention Networks (GATs) to identify informative prefrontal sensory connections associated with sleep-deprivation-induced network instability currently at the preprocessing and parcellation stage.

Adaptive Confidence Modeling for SHADER-Based 3D Human Body Shape Estimation

Ongoing

Python · PyTorch · OpenCV · SMPL

- Introduced a dynamic, image-driven confidence model replacing the original global- σ formulation in the SHADER monocular 3D shape estimation pipeline for loose-clothing scenarios.
- Achieved 20–25% lower confidence-prediction error on 9,656 dressed-human images ($p < 0.000001$), producing more stable SMPL estimation without additional data or architectural changes.

PETZZO : Comprehensive Pet Care Platform

Completed

Django · HTML · CSS · JavaScript · MySQL

- Built a full-stack platform connecting pet owners with verified hospitals, veterinarians, and pet shops, featuring AI-driven disease prediction, real-time pet tracking, and appointment booking.
- Implemented pet health record and vaccination management, pet adoption workflows, and streamlined pet shop operations within a single integrated system.

PM2.5 Air Quality Forecasting

Completed

Python · TensorFlow · LSTM · Pandas · Seaborn

- Developed an LSTM model forecasting 7-day PM2.5 levels across four major Indian cities; applied imputation, normalisation, and feature encoding with EDA visualisation.

Manglish WhatsApp Chatbot (Llama-3 / Ollama)

Completed

Node.js · Llama-3 · Ollama · Hybrid AI

- Built an offline Malayalam English chatbot combining rule-based intent analysis, similarity-based response retrieval, and local Llama-3 inference enabling cloud-independent real-time regional language messaging.